```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from imblearn.over_sampling import SMOTE

# 1. Collect Dataset
url = "https://storage.googleapis.com/download.tensorflow.org/data/creditcard.csv"
df = pd.read_csv(url)

print(f"Dataset Shape: {df.shape}")
print(df.head())
```

```
Dataset Shape: (284807, 31)
   Time        V1        V2        V3        V4        V5        V6        V7  \
0   0.0 -1.359807 -0.072781  2.536347  1.378155 -0.338321  0.462388  0.239599
1   0.0  1.191857  0.266151  0.166480  0.448154  0.060018 -0.082361 -0.078803
2   1.0 -1.358354 -1.340163  1.773209  0.379780 -0.503198  1.800499  0.791461
3   1.0 -0.966272 -0.185226  1.792993 -0.863291 -0.010309  1.247203  0.237609
4   2.0 -1.158233  0.877737  1.548718  0.403034 -0.407193  0.095921  0.592941

         V8        V9  ...       V21       V22       V23       V24       V25  \
0  0.098698  0.363787  ... -0.018307  0.277838 -0.110474  0.066928  0.128539
1  0.085102 -0.255425  ... -0.225775 -0.638672  0.101288 -0.339846  0.167170
2  0.247676 -1.514654  ...  0.247998  0.771679  0.909412 -0.689281 -0.327642
3  0.377436 -1.387024  ... -0.108300  0.005274 -0.190321 -1.175575  0.647376
4 -0.270533  0.817739  ... -0.009431  0.798278 -0.137458  0.141267 -0.206010

        V26       V27       V28  Amount  Class
0 -0.189115  0.133558 -0.021053  149.62      0
1  0.125895 -0.008983  0.014724    2.69      0
2 -0.139097 -0.055353 -0.059752  378.66      0
3 -0.221929  0.062723  0.061458  123.50      0
4  0.502292  0.219422  0.215153   69.99      0

[5 rows x 31 columns]
```

```
# 2. Preprocessing
# Scaling the 'Amount' and 'Time' columns
scaler = StandardScaler()
df['Amount'] = scaler.fit_transform(df['Amount'].values.reshape(-1, 1))
df['Time'] = scaler.fit_transform(df['Time'].values.reshape(-1, 1))

# 3. Feature Extraction (Defining X and y)
X = df.drop('Class', axis=1) # Features
y = df['Class']              # Target (0 = Normal, 1 = Fraud)

# Splitting into Training and Testing sets (80/20)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Handling Imbalance with SMOTE (Synthetic Minority Over-sampling Technique)
sm = SMOTE(random_state=42)
X_train_res, y_train_res = sm.fit_resample(X_train, y_train)

print(f"Original shape: {X_train.shape}")
print(f"Resampled shape: {X_train_res.shape}")
```

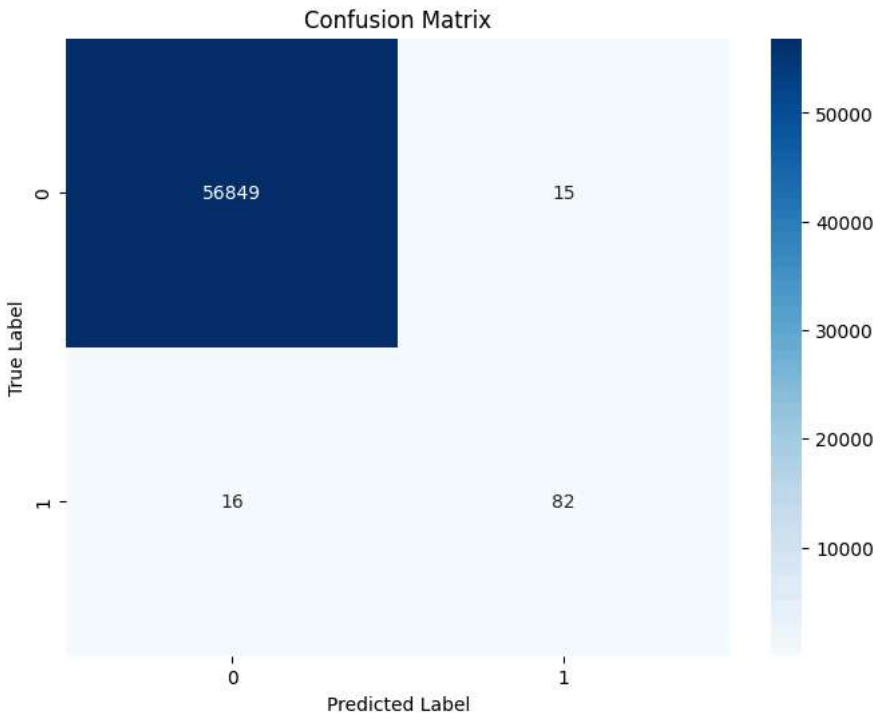
```
Original shape: (227845, 30)
Resampled shape: (454902, 30)
```

```
# 4. Training
print("Training the Random Forest model... please wait.")
model = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
model.fit(X_train_res, y_train_res)

# 5. Testing
y_pred = model.predict(X_test)
print("Testing complete.")
```

```
Training the Random Forest model... please wait.
Testing complete.
```

```
# 6. Evaluation
print("--- Model Evaluation ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Visualizing the Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues')
plt.title('Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()
```

```
--- Model Evaluation ---
Accuracy Score: 0.9995

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56864
           1       0.85      0.84      0.84        98

    accuracy                           1.00     56962
   macro avg       0.92      0.92      0.92     56962
weighted avg       1.00      1.00      1.00     56962
```



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